

Assignment: Practicing MongoDB CRUD & Aggregation

Inside it, create a collection named employees.

```
> use employee
< switched to db employee
> db.createCollection("faculty")
< { ok: 1 }
```

Perform MongoDB Tasks Step by Step

Task 1: Insert Documents

Insert at least 7 employee records with the following fields:

```
< { ok: 1 }
> db.faculty.insertMany([
  {eid:101,name:'yash diwate',age:30,department:'IT',Designation:'software engineer',salary:30000,city:'Mumbai'}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('68e6a0739598dd1b7c14fa00')
  }
}
> db.faculty.insertMany([
  {eid:102,name:'sarfaraz',age:23,department:'cs',Designation:'software engineer',salary:25000,city:'bangalore'},
  {eid:103,name:'irfan',age:25,department:'IT',Designation:'software engineer',salary:28000,city:'Mumbai'},
  {eid:104,name:'aditya',age:22,department:'IT',Designation:'software engineer',salary:26000,city:'Mumbai'},
  {eid:105,name:'priya',age:24,department:'IT',Designation:'software engineer',salary:27000,city:'Mumbai'},
  {eid:106,name:'rahul',age:26,department:'IT',Designation:'software engineer',salary:29000,city:'Mumbai'},
  {eid:107,name:'sneha',age:23,department:'IT',Designation:'software engineer',salary:25500,city:'Mumbai'},
  {eid:108,name:'vishal',age:27,department:'IT',Designation:'software engineer',salary:31000,city:'Mumbai'}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('68e6a69b9598dd1b7c14fa01'),
    '1': ObjectId('68e6a69b9598dd1b7c14fa02'),
    '2': ObjectId('68e6a69b9598dd1b7c14fa03'),
    '3': ObjectId('68e6a69b9598dd1b7c14fa04'),
    '4': ObjectId('68e6a69b9598dd1b7c14fa05'),
    '5': ObjectId('68e6a69b9598dd1b7c14fa06'),
    '6': ObjectId('68e6a69b9598dd1b7c14fa07')
  }
}
```

Task 2: Basic Queries

Find all employees.

```
}
> db.faculty.find()
< {
  _id: ObjectId('68e6a0739598dd1b7c14fa00'),
  eid: 101,
  name: 'yash diwate',
  age: 30,
  department: 'IT',
  Designation: 'software engineer',
  salary: 30000,
  city: 'Mumbai'
}
{
  _id: ObjectId('68e6a69b9598dd1b7c14fa01'),
  eid: 101,
  name: 'yash diwate',
  age: 30,
  department: 'IT',
  Designation: 'software engineer',
  salary: 30000,
  city: 'Mumbai'
}
{
```

Find employees who work in the "IT" department.

```
> db.faculty.find({department:'IT'})
< {
  _id: ObjectId('68e6a0739598dd1b7c14fa00'),
  eid: 101,
  name: 'yash diwate',
  age: 30,
  department: 'IT',
  Designation: 'software engineer',
  salary: 30000,
  city: 'Mumbai'
}
{
  _id: ObjectId('68e6a69b9598dd1b7c14fa01'),
  eid: 101,
  name: 'yash diwate',
  age: 30,
  department: 'IT',
  Designation: 'software engineer',
  salary: 30000,
  city: 'Mumbai'
}
```

Find employees with salary greater than 50,000.

```
> db.faculty.find({salary:{$gt:50000}})
< {
  _id: ObjectId('68e6a69b9598dd1b7c14fa04'),
  eid: 104,
  name: 'shaheen',
  age: 20,
  department: 'HR',
  Designation: 'chartered account',
  salary: 55000,
  city: 'lucknow'
}
{
  _id: ObjectId('68e6a69b9598dd1b7c14fa06'),
  eid: 106,
  name: 'abuzar',
  age: 28,
  department: 'BMS',
  Designation: 'senior developer',
  salary: 60000,
  city: 'indore'
}
```

Find one employee from "HR" department.

```
> db.faculty.find({department:'HR'})
< {
  _id: ObjectId('68e6a69b9598dd1b7c14fa04'),
  eid: 104,
  name: 'shaheen',
  age: 20,
  department: 'HR',
  Designation: 'chartered account',
  salary: 55000,
  city: 'lucknow'
}
```

Task 3:

Updates Update salary of "Yash Diwate" to 60000

```
> db.faculty.updateOne({name:'yash diwate'},{$set:{salary:60000}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.faculty.findOne({name:'yash diwate'})
< {
  _id: ObjectId('68e6a0739598dd1b7c14fa00'),
  eid: 101,
  name: 'yash diwate',
  age: 30,
  department: 'IT',
  Designation: 'software engineer',
  salary: 60000,
  city: 'Mumbai'
}
```

Change designation of "irfan" to "Senior Developer"

```
> db.faculty.updateOne({name:'irfan'},{$set:{Designation:'senior devloper'}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.faculty.findOne({name:'irfan'})
< {
  _id: ObjectId('68e6a69b9598dd1b7c14fa03'),
  eid: 103,
  name: 'irfan',
  age: 22,
  department: 'BT',
  Designation: 'senior devloper',
  salary: 35000,
  city: 'pune'
}
```

Update city of "sarfaraz" to "Pune"

```
> db.faculty.updateOne({name:'sarfaraz'},{$set:{city:'pune'}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.faculty.findOne({name:'sarfaraz'})
< {
  _id: ObjectId('68e6a69b9598dd1b7c14fa02'),
  eid: 102,
  name: 'sarfaraz',
  age: 23,
  department: 'cs',
  Designation: 'software engineer',
  salary: 25000,
  city: 'pune'
}
```

Task 4: Deletions

Delete one employee with salary less than 30,000.

```
> db.faculty.deleteMany({salary:{$lte:30000}})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.faculty.find()
< {
  _id: ObjectId('68e6a0739598dd1b7c14fa00'),
  eid: 101,
  name: 'yash diwate',
  age: 30,
  department: 'IT',
  Designation: 'software engineer',
  salary: 60000,
  city: 'Mumbai'
}
{
  _id: ObjectId('68e6a69b9598dd1b7c14fa03'),
  eid: 103,
  name: 'irfan',
  age: 22,
  department: 'BT',
  Designation: 'senior devloper',
```

Delete all employees from "Intern" designation.

```
> db.faculty.deleteMany({Designatuon:'intern'})
< {
  acknowledged: true,
  deletedCount: 0
}
employee>
```

